Host USB

USB serial
Commands, then TLS after PROVISION / ENCRYPT / DECRYPT / MANAGE

USB

MCU NonSecure

USB serial

Command parser,
then TLS forward

TrustZone

MCU Secure

Credentials
· SAE CA — provision peer verify
· device cert — mTLS client identity

TLS 1.3 client
post-quantum KEM

TrustZone entry

SPI

Device store
Reads copy one field then wipe. Pairing private and signature key stay off pad, peer, and public-key paths.
Device key seals the KEM ciphertext and the PIN state, and is mixed into the password hash and the seed wrap. Empty if erased.

Device key	**Record header**	**Operational**	**Secrets**	**Provision log**
generated once		time floor, fill identity, pad length start and end cursors, half bases pairing public, peers, owner ML-KEM public key	password hash pairing private signature private key	append only, 8 fills fill, arm, pad length, ends

Tropic chip

User slots 0–511. Pads are sealed under the ML-KEM shared secret, not with the device key.

KEM ciphertext	**Pads A**	**Pads B**	**ML-KEM seed**	**PIN state**
slots 0–2 MCU-sealed	slots 3–255 ML-KEM shared secret	slots 256–509 ML-KEM shared secret	slot 510 PIN-wrapped	slot 511 MCU-sealed

Separate from the user slots

Monotonic counters	**ECC slot 0**	**Pairing slots 0–3**	**MAC-and-Destroy 0–13**
encrypt and decrypt start encrypt and decrypt end	P-256 uplink signature	factory slot 0 host publics in 1–3	seven PIN tries